

## VRSAC, Serbia

WMO: 131830

Lat: 45.147N

Lon: 21.310E

Elev: 84

StdP: 100.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.2      | -8.1 | -13.7                         | 1.2 | -9.7 | -10.7 | 1.5 | -6.9 | 19.2                  | 2.2  | 17.3 | 3.0  | 1.3                   | 40   | 0.501 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.2                   | 34.2            | 21.1 | 32.6 | 21.1 | 31.0 | 20.5 | 23.0                | 30.6 | 22.1 | 29.8 | 21.3 | 28.8 | 3.5                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.7                            | 15.6 | 26.2 | 19.8 | 14.6 | 25.1 | 18.9 | 13.8 | 24.2 | 68.3          | 30.5 | 65.0 | 29.8 | 62.2 | 28.9 | 29.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 15.5          | 13.5 | 11.6 | -16.4                      | 36.3 | 4.6                | 2.3 | -19.8                                              | 37.9 | -22.5      | 39.3 | -25.1      | 40.5 | -28.5      | 42.2 |    |    |
| (5)               |      |      | -16.6                      | 24.6 | 4.6                | 1.5 | -19.9                                              | 25.7 | -22.7      | 26.6 | -25.3      | 27.4 | -28.6      | 28.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.5        | 1.2   | 3.4   | 7.4   | 12.9  | 17.5  | 20.7  | 22.5  | 22.8  | 18.0  | 12.7  | 8.0   | 2.5   |  |  |
|                                                                        | DBStd       | 8.87        | 5.55  | 5.69  | 5.00  | 4.38  | 3.99  | 3.79  | 3.51  | 3.78  | 4.17  | 4.82  | 5.32  | 5.18  |  |  |
|                                                                        | HDD10.0     | 966         | 276   | 193   | 111   | 21    | 1     | 0     | 0     | 0     | 0     | 29    | 99    | 237   |  |  |
|                                                                        | HDD18.3     | 2587        | 531   | 418   | 339   | 168   | 64    | 19    | 5     | 5     | 56    | 182   | 310   | 490   |  |  |
|                                                                        | CDD10.0     | 1888        | 3     | 8     | 30    | 109   | 233   | 321   | 387   | 398   | 240   | 113   | 41    | 6     |  |  |
|                                                                        | CDD18.3     | 467         | 0     | 0     | 1     | 6     | 38    | 89    | 134   | 144   | 45    | 8     | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 5154        | 0     | 0     | 1     | 53    | 334   | 991   | 1598  | 1699  | 438   | 39    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 2003        | 0     | 0     | 0     | 8     | 62    | 356   | 689   | 754   | 130   | 3     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.6         | 3.9   | 4.2   | 3.9   | 4.1   | 3.2   | 2.7   | 2.5   | 2.9   | 3.2   | 4.0   | 4.8   | 3.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 687         | 42    | 41    | 41    | 58    | 68    | 81    | 83    | 59    | 64    | 53    | 48    | 50    |  |  |
|                                                                        | PrecMax     | 941         | 85    | 85    | 106   | 123   | 124   | 189   | 246   | 186   | 185   | 127   | 133   | 133   |  |  |
|                                                                        | PrecMin     | 357         | 10    | 1     | 9     | 4     | 22    | 14    | 3     | 2     | 8     | 1     | 0     | 8     |  |  |
|                                                                        | PrecStd     | 146         | 21    | 24    | 25    | 29    | 28    | 44    | 60    | 41    | 39    | 34    | 27    | 31    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 14.9  | 18.2  | 22.3  | 27.8  | 30.2  | 34.1  | 36.3  | 37.0  | 33.1  | 26.6  | 22.2  | 15.9  |  |  |
|                                                                        |             | MCWB        | 9.7   | 10.7  | 12.6  | 16.5  | 19.3  | 21.7  | 21.5  | 21.2  | 19.7  | 17.4  | 14.5  | 10.7  |  |  |
|                                                                        | 2%          | DB          | 12.9  | 15.2  | 20.0  | 24.8  | 28.2  | 32.1  | 34.1  | 34.7  | 30.0  | 24.2  | 19.8  | 14.0  |  |  |
|                                                                        |             | MCWB        | 8.6   | 9.8   | 11.8  | 15.2  | 18.6  | 21.6  | 21.2  | 21.0  | 18.9  | 16.2  | 13.4  | 9.7   |  |  |
|                                                                        | 5%          | DB          | 11.1  | 13.4  | 17.8  | 22.5  | 26.9  | 30.6  | 32.2  | 32.8  | 27.9  | 22.2  | 17.8  | 12.1  |  |  |
|                                                                        |             | MCWB        | 7.6   | 8.6   | 10.7  | 14.0  | 17.9  | 20.9  | 21.2  | 20.8  | 18.0  | 15.4  | 12.4  | 8.3   |  |  |
|                                                                        | 10%         | DB          | 9.3   | 11.4  | 15.6  | 20.6  | 25.2  | 28.8  | 30.5  | 30.9  | 25.9  | 20.5  | 16.0  | 10.2  |  |  |
|                                                                        |             | MCWB        | 6.3   | 7.5   | 9.7   | 13.0  | 17.1  | 20.3  | 20.7  | 20.2  | 17.4  | 14.8  | 11.6  | 7.1   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.2  | 11.6  | 13.6  | 17.3  | 21.0  | 24.4  | 24.0  | 24.4  | 20.9  | 18.5  | 15.3  | 12.0  |  |  |
|                                                                        |             | MCDB        | 13.9  | 16.9  | 20.3  | 25.4  | 27.6  | 30.7  | 31.3  | 31.4  | 29.3  | 24.7  | 20.4  | 14.6  |  |  |
|                                                                        | 2%          | WB          | 9.0   | 10.2  | 12.4  | 15.8  | 19.7  | 22.7  | 22.9  | 22.7  | 19.7  | 17.2  | 13.9  | 10.0  |  |  |
|                                                                        |             | MCDB        | 12.4  | 14.4  | 18.4  | 23.1  | 26.2  | 29.8  | 30.5  | 30.5  | 27.2  | 22.3  | 18.6  | 13.5  |  |  |
|                                                                        | 5%          | WB          | 7.8   | 9.0   | 11.2  | 14.8  | 18.7  | 21.6  | 22.1  | 21.7  | 18.8  | 16.2  | 12.8  | 8.6   |  |  |
|                                                                        |             | MCDB        | 10.8  | 12.9  | 16.8  | 21.1  | 25.1  | 28.7  | 30.0  | 29.4  | 25.6  | 21.0  | 17.0  | 11.7  |  |  |
|                                                                        | 10%         | WB          | 6.6   | 7.9   | 10.1  | 13.7  | 17.7  | 20.7  | 21.3  | 20.9  | 18.0  | 15.2  | 11.7  | 7.4   |  |  |
|                                                                        |             | MCDB        | 9.0   | 11.0  | 14.7  | 19.3  | 24.0  | 27.4  | 28.7  | 28.7  | 24.2  | 20.0  | 15.7  | 9.9   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.2   | 7.6   | 9.7   | 10.2  | 10.8  | 11.5  | 12.3  | 12.2  | 10.7  | 9.7   | 7.6   | 6.0   |  |  |
|                                                                        |             | MCDBR       | 8.1   | 10.4  | 12.6  | 13.2  | 12.9  | 13.9  | 14.6  | 14.4  | 13.6  | 12.3  | 9.7   | 7.9   |  |  |
|                                                                        | 5% WB       | MCWBR       | 5.3   | 6.3   | 6.8   | 6.5   | 6.0   | 6.0   | 5.5   | 5.2   | 5.6   | 6.4   | 5.5   | 5.0   |  |  |
|                                                                        |             | MCWBR       | 7.4   | 9.6   | 11.3  | 12.1  | 12.0  | 13.1  | 12.7  | 12.7  | 11.8  | 11.1  | 9.0   | 7.6   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.344 | 0.371 | 0.406 | 0.461 | 0.448 | 0.441 | 0.449 | 0.462 | 0.435 | 0.395 | 0.361 | 0.342 |  |  |
|                                                                        |             | taud        | 2.320 | 2.238 | 2.197 | 2.057 | 2.161 | 2.217 | 2.194 | 2.157 | 2.199 | 2.301 | 2.349 | 2.347 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 746   | 793   | 816   | 799   | 826   | 832   | 819   | 791   | 777   | 757   | 719   | 707   |  |  |
|                                                                        |             | Edh at Noon | 81    | 105   | 125   | 157   | 146   | 139   | 140   | 141   | 124   | 98    | 79    | 72    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.20  | 2.04  | 3.29  | 4.58  | 5.56  | 6.16  | 6.19  | 5.51  | 3.85  | 2.63  | 1.58  | 0.99  |  |  |
|                                                                        | RadStd      | RadStd      | 0.14  | 0.27  | 0.42  | 0.46  | 0.53  | 0.56  | 0.47  | 0.52  | 0.42  | 0.29  | 0.14  | 0.14  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (61 neighbors) | +0.80 | N/A     | N/A    | +1.44   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|      |                         | +0.67 | N/A     | N/A    | N/A     | +0.38 | N/A   | -111        | -173    | +118    | +64     |

Nomenclature: See separate page